

Project Report — Advanced AI Medical Intelligence Platform

****Version:**** 1.0

****Date:**** 22 July 2026

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****Domain:**** Medical Image Analysis (Chest X-Ray Disease Classification)

1. Overview

End-to-end platform that:

1. Classifies chest X-rays as ****Normal**** or ****Pneumonia**** with a ResNet18 CNN.
2. Explains predictions with ****Grad-CAM**** heatmaps.
3. Generates an assistive narrative via an ****LLM**** (or template fallback).
4. Exposes functionality through a ****FastAPI**** REST API.
5. Persists history in ****PostgreSQL/SQLite****.
6. Deploys via ****Docker Compose**** with a React UI.

This is an educational demonstration, not a medical device. Every report includes a clinical disclaimer.

2. Architecture

Client (React / Streamlit)

| HTTPS REST/JSON

FastAPI (Uvicorn)

■■■ Inference Service (ResNet18)

■■■ Grad-CAM XAI Service

■■■ LLM Reporting Service

■■■ Persistence (SQLAlchemy → PostgreSQL/SQLite)

Cross-cutting: Docker Compose, `.env` config, CORS, structured logging with request IDs, static volumes for uploads/heatmaps.

3. Technology Stack

| Area | Technologies |

|-----|-----|

| Deep Learning | PyTorch, torchvision, ResNet18, Pillow, OpenCV, scikit-learn |

| XAI | pytorch-grad-cam, manual Grad-CAM hooks |

| LLM | OpenAI API (gpt-4o-mini), template fallback |

| API | FastAPI, Uvicorn, Pydantic, python-multipart |

Database	PostgreSQL / SQLite, SQLAlchemy, Alembic
Frontend	React, Vite, Tailwind, Axios, Recharts; Streamlit alternative
DevOps	Docker, Docker Compose, nginx
Quality	pytest, Black, Ruff

4. Database

``predictions``: id, image_filename, predicted_label, confidence_score, heatmap_path, llm_report, model_version, created_at, user_id (nullable).

Optional ``users`` table for future auth.

5. API Design

- ``GET /health``
- ``POST /api/predict``
- ``GET /api/history``, ``GET /api/history/{id}``, ``DELETE /api/history/{id}``
- ``GET /api/report/{id}?regenerate=``

Interactive docs: ``/docs``, ``/redoc``.

6. Grad-CAM & LLM

Grad-CAM uses the last ResNet18 convolutional block (``layer4[-1]``), overlays a jet colormap, and saves PNGs under ``static/heatmaps/``.

LLM prompts inject label, confidence, and region description; temperature ≤ 0.3 ; disclaimer always appended.

7. How to Run

See root ``README.md`` for Docker and local instructions.

8. Deliverables

- Source: ``backend/``, ``frontend/``, ``frontend_streamlit/``, ``model/``
- Checkpoint: ``model/checkpoints/best_model.pt``
- Docker: ``Dockerfile`s` + ``docker-compose.yml``
- Docs: this report + architecture notes

9. Disclaimer

This report and the associated software are AI-assisted technical demonstrations for educational/evaluation purposes only. They are not medical diagnoses and must not guide clinical care without licensed clinician review.